

Homework-2

1. Let $F \subseteq E$ be fields and $\alpha \in E$. Let e_α be the evaluation homomorphism

$$e_\alpha: F[x] \rightarrow E; f(x) \mapsto f(\alpha).$$

Clearly $e_\alpha(F[x]) = F[\alpha]$.

Prove that $\alpha \in E$ is transcendental

(i.e. not algebraic) over $F \Leftrightarrow F[\alpha] \cong F[x]$.

2. Let $F \subseteq E$ be fields, $\alpha, \beta \in E$. Suppose α is transcendental over F , but algebraic over $F(\beta)$. Show that β is algebraic over $F(\alpha)$.

Notation: $F(\alpha)$ for $\alpha \in E$ is given by

$$F(\alpha) = \left\{ \frac{f(\alpha)}{g(\alpha)} \mid f(x), g(x) \in F[x], g(\alpha) \neq 0 \right\}.$$

3. Let F be a finite field, i.e. $|F| < \infty$.
Prove that $|F| = p^n$ for some prime p & $n \in \mathbb{N}$.

4. Prove that every finite subgroup of $(\mathbb{C}^\times, \cdot)$ is cyclic.

5. Let $F \subseteq E$ be fields and $\alpha \in E$ be transcendental over F . Prove that every element of $F(\alpha)$ that is not in F , is also transcendental $/F$.
6. Let F be a finite field, $E \supseteq F$ an extension of fields, $\alpha \in E$ algebraic $/F$. Show that $|F(\alpha)| = q^n$, for $q = |F|$, for some $n \in \mathbb{N}$.
7. Let F be a finite field of characteristic p . Show that every element of F is algebraic over $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z} \subseteq F$.
8. Show that there exists a degree 3 irreducible polynomial over $\mathbb{F}_3 := \mathbb{Z}/3\mathbb{Z}$.
9. Construct a finite field having 27 elements.
10. Let p be a prime $\neq 2$. Prove that every element of $\mathbb{Z}/p\mathbb{Z}$ is a square of some element of $\mathbb{Z}/p\mathbb{Z}$.
11. Prove that $\exists$ finite fields having p^2 elements for every prime p .

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